(Ch #2)

Digital & Logic Design

Number System:-

Number system is simply a manner of counting. There are four number systems:

- Binary number system
- Octel number system
- Hexadecimal number system
- Decimal number system

Conversion of ON to BN system.

for example:

Truth table

Octal number	Binary number
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

$(1073)_8$

1	0	7	3
001	000	111	011

$(001000111011)_2$

B.N → O.N

001000111011

001 000 111 011

1 0 7 3

$(1073)_8$

Hexadecimal to binary:

H.N	B.N
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
A	1010
B	1011
C	1100
D	1101
E	1110
F	1111

for example

H.N → B.N

$(F9)_{16}$

= F 9

1111 1001

$(11111001)_2$

B.N → H.N

for example

$(0001110010000101)_2$

0001 1100 1000 0101

$(1C85)_{16}$

fractional form:-

$(AB.CD)_{16}$ H.N → B.N

$= (10101011.11001101)_2$

B.N → H.N

$(10.10)_2 = (0010.1000)_2$

$= (2.8)_2$

(Binary, Octal / Hexadecimal) → Decimal numbers

General rules:-

most significance bit MSB

low significance bit LSM

--- D D D D . D D ...

3 2 1 0 -1 -2

Base point

.... 3, 2, 1, 0, -1, -2, -3, ...

Integral form:-

for example:-

1) $(101)_2 = 1\times2^2+0\times2^1+1\times2^0$

$= 4+0+1$

$= (5)_{10}$

2) $(246)_8 = 2\times8^2+4\times8^1+6\times8^0$

$= 128+32+6$

$= (166)_{10}$

3) $(5A6)_{16} = 5\times16^2+10\times16^1+6\times16^0$

$= 1280+160+6$

$= (1446)_{10}$

Fractional form:-

$(101\cdot11)_2 = 1\times2^2+0\times2^1+1\times2^0+1\times2^{-1}+1\times2^{-2}$

$= 4+0+1+0\cdot5+0\cdot25$

$= (5\cdot75)_{10}$

$(5A6\cdot D1)_{16} =$

$= 5\times16^2+10\times16^1+6\times16^0+13\times16^{-1}+1\times16^{-2}$

$= 1280+160+6+0\cdot8125+0\cdot0039$

$= (1446\cdot8164)_{10}$

$(246\cdot7)_8 = ?$

$= 2\times8^2+4\times8^1+6\times8^0+7\times8^{-1}$

$= 128+32+6+0\cdot875$

$= (166\cdot875)_{10}$

Decimal to (Others)

$(51)_{10} = ?$

$= (0110011)_2$

		Remainder
2	51	
2	25	1
2	12	1
2	6	0
2	3	0
2	1	1
	0	1

$(51)_{10}$ to Octal

		Remainder
8	51	
8	6	3
8	0	6

$(51)_{10} = (063)_8$

$(51)_{10}$ to Hexadecimal

		Remainder
16	51	
16	3	3
	0	3

$= (033)_{16}$

Fractional form

Decimal to (Hexa/Binary/Octal)

$(0\cdot76)_{10} \longrightarrow$ Binary

	Result	fractional Part	integral
$0\cdot76\times2$	1·52	0·52	1
$0\cdot52\times2$	1·04	·04	1
$0\cdot04\times2$	0·08	·08	0
$0\cdot08\times2$	0·16	·16	0
$0\cdot16\times2$	0·32	·32	0

$(0\cdot11000)_2$

$(0.78)_{10} \rightarrow$ Octal

	Result	fractional	Integer
0.78×8	6.24	.24	6
0.24×8	1.92	0.92	1
0.92×8	7.36	0.36	7
0.36×8	2.88	0.88	2
0.88×8	7.04	.04	7

$(0.617)_8$

$(0.78)_{10} \rightarrow$ Hexadecimal

	Result	fractional	Integer
0.78×16	12.48	0.48	12
0.48×16	7.68	0.68	7
0.68×16	10.88	0.88	10
0.88×16	14.08	0.08	14
0.08×16	1.28	0.28	1

$(0.12710)_{16}$

Hexadecimal $\rightarrow$ Deci-

$(F16)_{16} \rightarrow$ Octal

$= 16 \times 16^2 + 1 \times 16^1 + 6 \times 16^0$

$= 4096 + 16 + 6$

$(4118)_{10}$

$=$

8	4118	
8	514	6
8	64	2
8	9	1
8	1	1
	0	1

$(011126)_8$

Convert $(25.58)_{10} \rightarrow$ Binary form.

2	25	
2	12	1
2	6	0
2	3	0
2	1	1
	0	1

011001

0.58×2	1.16	.16	1
0.16×2	0.32	.32	0
0.32×2	0.64	.64	0
0.64×2	1.28	0.28	1
.28×2	0.56	.56	0

$(011001.10010)_2$

25 . 0.58

Binary Arithmetic

1 1 0 1 1
+ 1 0 0 1 1
$(101110)_2$

1 0
+ 1 1
$(101)_2$

1 1 1
+ 0 1 1
$(1010)_2$

Subtraction

1 1 1
− 1 0 1
$(010)_2$

1 1 0
− 1 0 1
$(001)_2$

0 1 1
− 1 1 1
$(-100)_2$

Binary Multiplication

1 1 1 0
× 1 0 1
1 1 1 0
0 0 0 0 X
1 1 1 0 X X
$(1000110)_2$

1 1 . 1 0
× 1 . 1
1 1 1 0
1 1 1 0 X
$(101.010)_2$

1 1 1 1
× 0 1 1 1
1 1 1 1
1 1 1 1 X
1 1 1 1 X X
0 0 0 0 X X X
$(1101001)_2$

1 1 1
× 1 0 1
1 1 1
0 0 0 X
1 1 1 X X
$(100011)_2$

1 0 1 0 1
× 1 0 1
1 0 1 0 1
0 0 0 0 0 X
1 0 1 0 1 X X
$(1101001)_2$

1 0 1 1 1
× 1 0 1
1 0 1 1 1
0 0 0 0 0 X
1 0 1 1 1 X X
$(1011001 1)_2$

Binary Division

$(101010)_2 \div$ by $(110)_2$

$(000111)_2$ Ans

```
       000111
110 √ 101010
      -0↓
      10
      00
      101
       0
      1010
       110
      1001
       110
       110
       110
        x
```

$(111000)_2 \div$ by $(111)_2$

```
       1000
111 √ 111000
      111
      000 → Remainder
```

$(1000)_2$ Ans

$(1110111)_2 \div$ by $(1001)_2$

$(1101)_2$ → Ans

```
        1101
1001 √ 1110111
       1001
       1011
       1001
        101
          0
       1011
       1001
        110
```

$(00101101)_2 \div$ by $(11)_2$

```
       01111
11 √ 00101101
       01
       101
        11
       101
        11
       100
        11
        11
        11
         x
```

$(100101)_2 \div$ by $(101)_2$

$(00111)_2 \rightarrow$ Ans

```
        00111
101 ) 100101
     -0
      100
        0
      1001
       101
      1000
       101
       111
       101
       010
```

$11011 \div 100$

```
          11
100 ) 11011
      100
       101
       100
        11
```

$1100010 \div 111$

```
         00011
111 ) 1100010
     -0
      11
       0
      110
        0
      1100
       111
      1110
       111
        01
         0
```

Finding the 1's Complement:

(a) $(10110010)_2$

After 1's complement

$= (01001101)_2$

(b) $(10110010)_2$

After 1's complement

$= (01001101)_2$

(c) $(10111000)_2$

After 1's complement

$= (01000111)_2$.

Finding the 2's complement:

Find the 2's complement of 1011001o.

After 1's complement

0100110 1

2's complement = 1's + 1

= 0100 1101

+ 1

01001110

Find the 2's complement of 10111000 using the alternative method.

= 10111 000 Binary number
= (01001 000)₂ 2's complement

1's complements of original bits — These bits stay the same

a) 00010110

After 2's complement

11101010

b) 1111100

After 2's complement

c)

00000100

c) 10010001

After 2's complement

01101111

Subtraction Using 1's complement

Q#1 Solve $(1101)_2 - (1001)_2$ Using 1's complement.

$P = (1101)_2$

$Q = (1001)_2$

$-Q = $ 1's comp $= (0110)_2$

$$
\begin{array}{rrrrr}
P+(-Q) = & 1 & 1 & 0 & 1 \\
+ & 0 & 1 & 1 & 0 \\
\hline
\text{End carry} \leftarrow \boxed{1} & 0 & 0 & 1 & 1 \\
+ & & & & 1 \\
\hline
& 0 & 1 & 0 & 0
\end{array}
$$

$(0100)_2$

Solve Using 1's complement.

$(1001)_2 - (1101)_2$

$P = (1001)_2$

$Q = (1101)_2$

$-Q = $ 1's complement $= (0010)_2$

$$
\begin{array}{rrrrr}
P+(-Q) = & 1 & 0 & 0 & 1 \\
+ & 0 & 0 & 1 & 0 \\
\hline
\text{No carry} \leftarrow & 1 & 0 & 1 & 1
\end{array}
\rightarrow \text{Again taking 1's complement}
$$

Q#2 $(0100)_2$ Ans

Subtracting complement of 2's

Solve

$(1111)_2 - (1101)_2$

P = $(1111)_2$

Q = $(1101)_2$

-Q = $(0010)_2$

P+(-Q) =

1 1 1 1
+ 0 0 1 0
$\boxed{1}$ 0 0 0 1

P = $(1111)_2$

Q = $(1101)_2$

-Q = After 2's complement = $(0011)_2$

P+(-Q) =

1 1 1 1
+ 0 0 1 1
Skipped end carry ← $\boxed{1}$ 0 0 1 0

$(0010)_2$ final result.

Solve using 2's complement.

$(1101)_2 - (1111)_2$

P = $(1101)_2$

Q = $(1111)_2$

-Q = After 2's complement = $(0000)_2$

P+(-Q) =

1 1 0 1
+ 0 0 0 0
1 1 0 1

After taking 2's complement

$(0010)_2$

Solve using 1's complement

$(1111)_2 - (1101)_2$

P = $(1111)_2$

Q = $(1101)_2$

-Q = After 1's complement = $(0010)_2$

P+(-Q) =

1 1 1 1
+ 0 0 1 0
end carry ← $\boxed{1}$ 0 0 0 1
+ 1
$(0010)_2$

$(0010)_2$

Using 1's complement

$(1101)_2 - (1111)_2$

$P = (1101)_2$

$Q = (1111)_2$

$-Q = (0000)_2$

$P + (-Q) =$

$$\begin{array}{r} 1\ 1\ 0\ 1 \\ +\ 0\ 0\ 0\ 0 \\ \hline 1\ 1\ 0\ 1 \end{array}$$

No carry ← 1 1 0 1

↓

Taking 1's complement again

$= (0010)_2$

Solve Using 2's complement.

$(1101)_2 - (1001)_2$

$P = (1101)_2$

$Q = (1001)_2$

$-Q =$ After 2's complement $= \cancel{(0110)_2}$ (0111)

$P + (-Q) =$

$$\begin{array}{r} 1\ 1\ 0\ 1 \\ +\ 0\ 1\ 1\ 1 \\ \hline \boxed{1}\ 0\ 1\ 0\ 0 \end{array}$$

Skipped end carry ← $\boxed{1}$ 0 1 0 0

$(0100)_2$ final result.